FACULTY OF COMPUTING & DATA SCIENCE

$1011_2 + 110_2$

$1776_8 - 345_8$

$DAD_{16} + EB5_{16}$

Binary Octal Hexadecimal

SECTION 4

ADDITION & SUBTRACTION

MIND MAP

TODAY ON ONE PAGE

BINARY base 2 · 0–1

1 + 1 = 10 · + carry at 2 · − borrow 2

OCTAL base 8 · 0–7

7 + 1 = 10 · + carry at 8 · − borrow 8

+ −

in 3 bases

HEX base 16 · 0–F

A–F = 10–15 · + carry at 16 · − borrow 16

ONE RULE

Work right to left. Reach the base → carry 1. Run short → borrow one base.

TODAY

THE PLAN

Learn the rule once, then apply it in three bases.

01	Bases & the one rule	foundations
02	Binary	+ -
03	Octal	+ -
04	Hexadecimal	+ -
05	Practice · Sheet 4	8 problems
06	Answer key · Sheet 3	32 answers

FOUNDATIONS

THREE BASES AT A GLANCE

Place values (right to left)

Binary	8 (2^3)	4 (2^2)	2 (2^1)	1 (2^0)
Octal	512 (8^3)	64 (8^2)	8 (8^1)	1 (8^0)
Hex	4096 (16^3)	256 (16^2)	16 (16^1)	1 (16^0)

Digits allowed: binary 0–1 · octal 0–7 · hex 0–9, A–F.

CHART

255 written in each base

Binary	11111111	8 digits
Octal	377	3 digits
Decimal	255	3 digits
Hex	FF	2 digits

Bigger base → fewer digits.

THE CORE IDEA

ONE RULE, EVERY BASE

OPERATION	BINARY · 2	OCTAL · 8	HEX · 16
Add	sum ≥ 2 → carry 1	sum ≥ 8 → carry 1	sum ≥ 16 → carry 1
Subtract	borrow adds 2	borrow adds 8	borrow adds 16
“10” means	two	eight	sixteen

Check any answer by converting both numbers to decimal.

01

base 2 · digits 0, 1

BINARY

Add · Subtract

BINARY · ADDITION

1 + 1 = 10

0 + 0 = 0

nothing to carry

0 + 1 = 1

nothing to carry

1 + 1 = 10

write 0, carry 1

1 + 1 + 1 = 11

write 1, carry 1

BINARY · ADDITION

CARRIES CHAIN ALONG

Example 1 / 3

SHEET 3 · Q1E

```
  1  1    1  1  1
    10111
+ 110001
--------
  1001000
```

≡ 23 + 49 = 72 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	1 + 1 = 2	0 (2 - 2)	1
2	1 + 0 + 1 = 2	0 (2 - 2)	1
3	1 + 0 + 1 = 2	0 (2 - 2)	1
4	0 + 0 + 1 = 1	1	-
5	1 + 1 = 2	0 (2 - 2)	1
6	1 + 1 = 2	0 (2 - 2)	1
	leftover carry	1	-

BINARY · ADDITION

CARRY OFF THE END

Example 2 / 3

SHEET 3 · Q1D

```
  1     1
  10001
+ 11101
-------
 101110
```

≡ 17 + 29 = 46 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	1 + 1 = 2	0 (2 - 2)	1
2	0 + 0 + 1 = 1	1	–
3	0 + 1 = 1	1	–
4	0 + 1 = 1	1	–
5	1 + 1 = 2	0 (2 - 2)	1
	leftover carry	1	–

BINARY · ADDITION

LINE UP THE POINTS

Example 3 / 3

SHEET 3 · Q1F

```
      1  1    1
   1001.011
+  0100.110
-----------
   1110.001
```

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	1 + 0 = 1	1	–
2	1 + 1 = 2	0 (2 - 2)	1
3	0 + 1 + 1 = 2	0 (2 - 2)	1
4	1 + 0 + 1 = 2	0 (2 - 2)	1
5	0 + 0 + 1 = 1	1	–
6	0 + 1 = 1	1	–
7	1 + 0 = 1	1	–

BINARY · SUBTRACTION

A BORROW IS WORTH 2

`0 - 0 = 0`

no borrow

`1 - 0 = 1`

no borrow

`1 - 1 = 0`

no borrow

`0 - 1 = 1`

borrow: 10 − 1 = 1

BINARY · SUBTRACTION

A BORROW IS WORTH 2

Example 1 / 3

SHEET 3 · Q2C

```
    0  2
   1 1 0 1 1
 - 1 0 1 1 0
 -----------
       1 0 1
```

≡ 27 − 22 = 5 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	1 - 0 = 1	1	–
2	1 - 1 = 0	0	–
3	0 + 2 - 1 = 1	1	yes
4	1 - 1 - 0 = 0	0	–
5	1 - 1 = 0	0	–

BINARY · SUBTRACTION

BORROWING TWICE IN A ROW

Example 2 / 3

SHEET 3 · Q2E

```
  0  2  1  2
 ~~1~~ ~~1~~ ~~0~~ 0
-  0  1  1  1
-------------
     1  0  1
```

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	0 + 2 - 1 = 1	1	yes
2	0 - 1 + 2 - 1 = 0	0	yes
3	1 - 1 + 2 - 1 = 1	1	yes
4	1 - 1 - 0 = 0	0	–

≡ 12 – 7 = 5 (decimal)

BINARY · SUBTRACTION

BORROW ONCE, THEN EASY

Example 3 / 3

SHEET 3 · Q2D

```
     0  2  2
   1 1 1 0
 - 0 0 1 1
 ---------
   1 0 1 1
```

≡ 14 – 3 = 11 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	0 + 2 - 1 = 1	1	yes
2	1 - 1 + 2 - 1 = 1	1	yes
3	1 - 1 - 0 = 0	0	–
4	1 - 0 = 1	1	–

02

base 8 · digits 0–7

OCTAL

8 and 9 don’t exist here

OCTAL · ADDITION

SUM ≥ 8 → CARRY 1

3 + 4 = 7

below 8: write it as is

7 + 1 = 10

8 = 1×8 + 0

5 + 4 = 11

9 = 1×8 + 1

6 + 7 = 15

13 = 1×8 + 5

OCTAL · ADDITION

SUM ≥ 8 → CARRY 1

Example 1 / 3

SHEET 3 · Q5D

```
    1 1 1
  1 7 7 6
+   3 4 5
-------
  2 3 4 3
```

≡ 1022 + 229 = 1251 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	6 + 5 = 11	3 (11 - 8)	1
2	7 + 4 + 1 = 12	4 (12 - 8)	1
3	7 + 3 + 1 = 11	3 (11 - 8)	1
4	1 + 1 = 2	2	–

OCTAL · ADDITION

A CARRY IN EVERY COLUMN

Example 2 / 3

SHEET 3 · Q5C

```
  1  1  1
    567
+   243
-------
   1032
```

≡ 375 + 163 = 538 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	7 + 3 = 10	2 (10 - 8)	1
2	6 + 4 + 1 = 11	3 (11 - 8)	1
3	5 + 2 + 1 = 8	0 (8 - 8)	1
	leftover carry	1	–

OCTAL · ADDITION

SAME RULE, WITH A POINT

Example 3 / 3

SHEET 3 · Q5F

```
  1  1  1
  366.23
+ 243.62
--------
  632.05
```

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	3 + 2 = 5	5	–
2	2 + 6 = 8	0 (8 - 8)	1
3	6 + 3 + 1 = 10	2 (10 - 8)	1
4	6 + 4 + 1 = 11	3 (11 - 8)	1
5	3 + 2 + 1 = 6	6	–

OCTAL · SUBTRACTION

A BORROW IS WORTH 8

6 - 2 = 4

no borrow

3 - 5 = 6

borrow: 11 – 5 = 6

0 - 1 = 7

borrow: 8 – 1 = 7

2 - 7 = 3

borrow: 10 – 7 = 3

OCTAL · SUBTRACTION

ZEROS PASS THE BORROW

Example 1 / 3

SHEET 3 · Q6B

```
    3  7  9
  2 4 0 1
- 1 0 7 7
-------
  1 3 0 2
```

≡ 1281 – 575 = 706 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	1 + 8 - 7 = 2	2	yes
2	0 - 1 + 8 - 7 = 0	0	yes
3	4 - 1 - 0 = 3	3	–
4	2 - 1 = 1	1	–

OCTAL · SUBTRACTION

A BORROW IS WORTH 8

Example 2 / 3

SHEET 3 · Q6A

```
   2  12 14
   3  5  6
 - 1  5  7
 ---------
   1  7  7
```

≡ 238 − 111 = 127 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	6 + 8 - 7 = 7	7	yes
2	5 - 1 + 8 - 5 = 7	7	yes
3	3 - 1 - 1 = 1	1	–

OCTAL · SUBTRACTION

ONE BORROW IN THE MIDDLE

Example 3 / 3

SHEET 3 · Q6C

```
   6  13
   ~~7~~52
 - 361
 -----
   371
```

≡ 490 – 241 = 249 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	2 - 1 = 1	1	–
2	5 + 8 - 6 = 7	7	yes
3	7 - 1 - 3 = 3	3	–

03

base 16 · digits 0–9, A–F

HEXADECIMAL

Letters are digits too

HEX · DIGIT TABLE

SIXTEEN DIGITS

0 0	1 1	2 2	3 3	4 4	5 5	6 6	7 7
8 8	9 9	A 10	B 11	C 12	D 13	E 14	F 15

HEX · ADDITION

SUM ≥ 16 → CARRY 1

8 + 7 = F

15 is below 16: write F

A + 6 = 10

16 = 1×16 + 0

D + 5 = 12

18 = 1×16 + 2

F + F = 1E

30 = 1×16 + 14

HEX · ADDITION

SUM ≥ 16 → CARRY 1

Example 1 / 3

SHEET 3 · Q8C

```
   1  1  1
     DAD
+    EB5
-------
   1C62
```

≡ 3501 + 3765 = 7266 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	13 + 5 = 18	2 (18 - 16)	1
2	10 + 11 + 1 = 22	6 (22 - 16)	1
3	13 + 14 + 1 = 28	C (28 - 16)	1
	leftover carry	1	-

HEX · ADDITION

LETTERS IN, LETTERS OUT

Example 2 / 3

SHEET 3 · Q8B

```
  1  1  1
    8AB
+   B78
-------
   1423
```

≡ 2219 + 2936 = 5155 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	11 + 8 = 19	3 (19 - 16)	1
2	10 + 7 + 1 = 18	2 (18 - 16)	1
3	8 + 11 + 1 = 20	4 (20 - 16)	1
	leftover carry	1	-

HEX · ADDITION

WORK IN DECIMAL, WRITE IN HEX

Example 3 / 3

SHEET 3 · Q8A

```
    1  1
    3F8
+   3BF
-------
   7B7
```

≡ 1016 + 959 = 1975 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	8 + 15 = 23	7 (23 - 16)	1
2	15 + 11 + 1 = 27	B (27 - 16)	1
3	3 + 3 + 1 = 7	7	–

HEX · SUBTRACTION

A BORROW IS WORTH 16

E - 5 = 9

14 − 5 = 9

B - C = F

borrow: 27 − 12 = 15

0 - 1 = F

borrow: 16 − 1 = 15

4 - 9 = B

borrow: 20 − 9 = 11

HEX · SUBTRACTION

A BORROW IS WORTH 16

Example 1 / 3

SHEET 3 · Q9B

```
   8  22 19 27
   9  7  4  B
 - 5  8  7  C
 ------------
   3  E  C  F
```

≡ 38731 – 22652 = 16079 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	11 + 16 - 12 = 15	F (15)	yes
2	4 - 1 + 16 - 7 = 12	C (12)	yes
3	7 - 1 + 16 - 8 = 14	E (14)	yes
4	9 - 1 - 5 = 3	3	–

HEX · SUBTRACTION

BORROW TWICE

Example 2 / 3

SHEET 3 · Q9C

```
    E 25 27
   F A B
 - A E F
 -------
   4 B C
```

≡ 4011 − 2799 = 1212 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	11 + 16 - 15 = 12	C (12)	yes
2	10 - 1 + 16 - 14 = 11	B (11)	yes
3	15 - 1 - 10 = 4	4	–

HEX · SUBTRACTION

NO BORROW NEEDED

Example 3 / 3

SHEET 3 · Q9A

```
  9654
- 5321
------
  4333
```

≡ 38484 – 21281 = 17203 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	4 - 1 = 3	3	–
2	5 - 2 = 3	3	–
3	6 - 3 = 3	3	–
4	9 - 5 = 4	4	–

WATCH OUT

COMMON MISTAKES

Carrying at 10

WRONG

~~$7_8 + 5_8 = 12$~~

RIGHT

$7_8 + 5_8 = 14$

12 = 1×8 + 4. Carry happens at the base.

Borrowing 10

WRONG

~~$10_8 - 7 = 3$~~

RIGHT

$10_8 - 7 = 1$

The borrowed 1 is worth 8, not 10: 8 − 7 = 1.

Two digits in a column

WRONG

~~B + 9 = 20~~

RIGHT

$B + 9 = 14_{16}$

20 = 1×16 + 4. Each column gets one digit.

Points misaligned

WRONG

~~53.012+742.2~~

RIGHT

053.012+742.200

Line up the points and pad with zeros first.

04

Sheet 4 · 8 problems

PRACTICE

Octal and hex, with full solutions

SHEET 4 · YOUR TURN

EIGHT PROBLEMS

Octal

Q1 512 + 723

Q2 720 + 236

Q3 751 - 224

Q4 6334 - 732

Hexadecimal

Q7 15FA + DAC

Q8 4E2C + 1395

Q9 F952 - 523

Q10 9523 - 24B

SHEET 4 · Q1 · YOUR TURN

OCTAL ADDITION

BASE 8

$$\begin{array}{r} 512 \\ +\ 723 \\ \hline \end{array}$$

HINT

0 1 2 3 4 5 6 7

Digits stop at 7. No 8, no 9.

Column sum ≥ 8? Subtract 8, write the rest, carry 1.

SHEET 4 · Q1 · OCTAL ADDITION

512 + 723

ANSWER

1435

```
   1
   512
+  723
------
  1435
```

≡ 330 + 467 = 797 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	2 + 3 = 5	5	–
2	1 + 2 = 3	3	–
3	5 + 7 = 12	4 (12 - 8)	1
	leftover carry	1	–

SHEET 4 · Q2 · YOUR TURN

OCTAL ADDITION

BASE 8

$$\begin{array}{r} 720 \\ +\ 236 \\ \hline \end{array}$$

HINT

0 1 2 3 4 5 6 7

Digits stop at 7. No 8, no 9.

Column sum ≥ 8? Subtract 8, write the rest, carry 1.

SHEET 4 · Q2 · OCTAL ADDITION

720 + 236

ANSWER

1156

```
   1
   720
+  236
------
  1156
```

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	0 + 6 = 6	6	–
2	2 + 3 = 5	5	–
3	7 + 2 = 9	1 (9 - 8)	1
	leftover carry	1	–

≡ 464 + 158 = 622 (decimal)

SHEET 4 · Q3 · YOUR TURN

OCTAL SUBTRACTION

BASE 8

$$\begin{array}{r} 751 \\ -\,224 \\ \hline \end{array}$$

HINT

0 1 2 3 4 5 6 7

Digits stop at 7. No 8, no 9.

Top digit too small? Borrow 1 from the left. It is worth 8.

SHEET 4 · Q3 · OCTAL SUBTRACTION

751 - 224

ANSWER

525

```
    4  9
  7 5 1
- 2 2 4
-------
  5 2 5
```

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	1 + 8 - 4 = 5	5	yes
2	5 - 1 - 2 = 2	2	–
3	7 - 2 = 5	5	–

≡ 489 – 148 = 341 (decimal)

SHEET 4 · Q4 · YOUR TURN

OCTAL SUBTRACTION

BASE 8

$$\begin{array}{r} 6334 \\ -\ 732 \\ \hline \end{array}$$

HINT

0 1 2 3 4 5 6 7

Digits stop at 7. No 8, no 9.

Top digit too small? Borrow 1 from the left. It is worth 8.

SHEET 4 · Q4 · OCTAL SUBTRACTION

6334 - 732

ANSWER

5402

```
   5  11
   6334
 -  732
 ------
   5402
```

≡ 3292 − 474 = 2818 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	4 - 2 = 2	2	–
2	3 - 3 = 0	0	–
3	3 + 8 - 7 = 4	4	yes
4	6 - 1 - 0 = 5	5	–

SHEET 4 · Q7 · YOUR TURN

HEX ADDITION

BASE 16

```
  15FA
+  DAC
------
```

HINT

0	1	2	3	4	5	6	7
8	9	A	B	C	D	E	F

A=10 … F=15. Nothing above F.

Column sum ≥ 16? Subtract 16, write the rest, carry 1.

SHEET 4 · Q7 · HEX ADDITION

15FA + DAC

ANSWER

23A6

```
   1  1  1
    15FA
+    DAC
------
    23A6
```

≡ 5626 + 3500 = 9126 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	10 + 12 = 22	6 (22 - 16)	1
2	15 + 10 + 1 = 26	A (26 - 16)	1
3	5 + 13 + 1 = 19	3 (19 - 16)	1
4	1 + 1 = 2	2	–

SHEET 4 · Q8 · YOUR TURN

HEX ADDITION

BASE 16

```
  4E2C
+ 1395
------
```

HINT

0 1 2 3 4 5 6 7
8 9 A B C D E F

A=10 … F=15. Nothing above F.

Column sum ≥ 16? Subtract 16, write the rest, carry 1.

SHEET 4 · Q8 · HEX ADDITION

4E2C + 1395

ANSWER

61C1

```
   1   1
   4E2C
+  1395
------
   61C1
```

≡ 20012 + 5013 = 25025 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	ADD	WRITE	CARRY
1	12 + 5 = 17	1 (17 - 16)	1
2	2 + 9 + 1 = 12	C (12)	–
3	14 + 3 = 17	1 (17 - 16)	1
4	4 + 1 + 1 = 6	6	–

SHEET 4 · Q9 · YOUR TURN

HEX SUBTRACTION BASE 16

$$\begin{array}{r} F952 \\ -\ \ 523 \\ \hline \end{array}$$

HINT

0	1	2	3	4	5	6	7
8	9	A	B	C	D	E	F

A=10 … F=15. Nothing above F.

Top digit too small? Borrow 1 from the left. It is worth 16.

SHEET 4 · Q9 · HEX SUBTRACTION

F952 - 523

ANSWER

F42F

```
     4 18
   F 9 5 2
 -   5 2 3
 ---------
   F 4 2 F
```

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	2 + 16 - 3 = 15	F (15)	yes
2	5 - 1 - 2 = 2	2	–
3	9 - 5 = 4	4	–
4	15 - 0 = 15	F (15)	–

≡ 63826 – 1315 = 62511 (decimal)

SHEET 4 · Q10 · YOUR TURN

HEX SUBTRACTION BASE 16

$$\begin{array}{r} 9523 \\ -\ 24B \\ \hline \end{array}$$

HINT

0	1	2	3	4	5	6	7
8	9	A	B	C	D	E	F

A=10 … F=15. Nothing above F.

Top digit too small? Borrow 1 from the left. It is worth 16.

SHEET 4 · Q10 · HEX SUBTRACTION

9523 - 24B

ANSWER

92D8

```
     4  17 19
    9 5 2 3
  -   2 4 B
  ---------
    9 2 D 8
```

≡ 38179 − 587 = 37592 (decimal)

RIGHT TO LEFT · ONE COLUMN PER ROW

COL	SUBTRACT	WRITE	BORROW?
1	3 + 16 - 11 = 8	8	yes
2	2 - 1 + 16 - 4 = 13	D (13)	yes
3	5 - 1 - 2 = 2	2	–
4	9 - 0 = 9	9	–

05

Sheet 3 · 32 answers

ANSWER KEY

Addition and subtraction only

SHEET 3 · ANSWER KEY 1 OF 3

BINARY

Addition		Q1
a	11 + 1	100
b	11 + 11	110
c	101 + 101	1010
d	10001 + 11101	101110
e	10111 + 110001	1001000
f	1001.011 + 0100.110	1110.001

Subtraction		Q2
a	11 - 01	10
b	101 - 011	010
c	11011 - 10110	00101
d	1110 - 0011	1011
e	1100 - 0111	0101

SHEET 3 · ANSWER KEY 2 OF 3

OCTAL

Addition		Q5
a	5 + 4	11
b	243 + 212	455
c	567 + 243	1032
d	1776 + 345	2343
e	147 + 261	430
f	366.23 + 243.62	632.05
g	53.012 + 742.2	1015.212

Subtraction		Q6
a	356 - 157	177
b	2401 - 1077	1302
c	752 - 361	371
d	1304007 - 35006	1247001
e	272.423 - 34.4444	235.7564
f	543.125 - 271.34	251.565

SHEET 3 · ANSWER KEY 3 OF 3

HEXADECIMAL

Addition	Q8
a 3F8 + 3BF	7B7
b 8AB + B78	1423
c DAD + EB5	1C62
d 98F.A2 + B11.94	14A1.36

Subtraction	Q9
a 9654 - 5321	4333
b 974B - 587C	3ECF
c FAB - AEF	4BC
d F3A.251 - FE.64	E3B.C11

SOLVE IT ON THE BOARD

BONUS CHALLENGE

+ bonus marks

Q1 · ANSWER

101000

Q2 · ANSWER

3235

Q3 · ANSWER

1070

Q4 · ANSWER

100E

BONUS CHALLENGE

SOLUTIONS

Q1 · BINARY

```
  1 1 1 1 1
   11011
+   1101
 ------
  101000
```

≡ 27 + 13 = 40 (decimal)

Q2 · OCTAL

```
   6 7 9 12
  ~~7~~02~~4~~
- 3567
 ------
  3235
```

≡ 3604 – 1911 = 1693 (decimal)

Q3 · HEX

```
  1 1 1
   9C7
+  6A9
 ------
  1070
```

≡ 2503 + 1705 = 4208 (decimal)

Q4 · HEX

```
    9 15 19
  1~~A~~0~~3~~
-  9F5
 ------
  100E
```

≡ 6659 – 2549 = 4110 (decimal)

SUMMARY

WHAT TO REMEMBER

+ ADD

Column sum ≥ base: write sum − base, carry 1.

- SUBTRACT

Top digit too small: borrow, which adds one base.

base b

2 · 8 · 16

. ALIGN

Line up the points. Pad empty places with 0.

= CHECK

Convert both numbers to decimal and compare.

Next time · Section 5: multiplication

ANY QUESTIONS?